

Problem (Problem 1): Let $\Gamma \curvearrowright L^\infty(X, \mu)$ for a countable discrete group Γ and a probability space (X, μ) .

- (a) For $f \in L^\infty(X, \mu)$, show that $\pi_\alpha(f)$ is a bounded operator on $\ell^2(\Gamma) \otimes L^2(X, \mu)$ with $\|\pi_\alpha(f)\| = \|f\|_{L^\infty}$.
- (b) Show that $\pi_\alpha: L^\infty(X, \mu) \rightarrow B(\ell^2(\Gamma) \otimes L^2(X, \mu))$ is a unital $*$ -homomorphism.
- (c) Show that $\lambda_g \pi_\alpha(f) \lambda_{g^{-1}} = \pi_\alpha(\alpha_g(f))$.

Solution:

- (a) We start by recalling the definition of $\pi_\alpha(f)$, given by action on elementary tensors

$$\pi_\alpha(f)(\delta_g \otimes f_g) = \delta_g \otimes \alpha_{g^{-1}}(f)f_g.$$

Computing norms, we have

$$\begin{aligned} \left\| \pi_\alpha(f) \left(\sum_{g \in \Gamma} \delta_g \otimes f_g \right) \right\|^2 &= \left\langle \pi_\alpha(f) \left(\sum_{g \in \Gamma} \delta_g \otimes f_g \right), \pi_\alpha(f) \left(\sum_{g \in \Gamma} \delta_g \otimes f_g \right) \right\rangle \\ &= \sum_{g, h \in \Gamma} \langle \delta_g \otimes \alpha_{g^{-1}}(f)f_g, \delta_h \otimes \alpha_{h^{-1}}(f)f_h \rangle \\ &= \sum_{g, h \in \Gamma} \langle \delta_g, \delta_h \rangle \langle \alpha_{g^{-1}}(f)f_g, \alpha_{h^{-1}}(f)f_h \rangle \\ &= \sum_{g \in \Gamma} \langle \alpha_{g^{-1}}(f)f_g, \alpha_{g^{-1}}(f)f_g \rangle \\ &\leq \sum_{g \in \Gamma} \|\alpha_{g^{-1}}(f)\|_{L^\infty}^2 \|f_g\|^2 \\ &= \sum_{g \in \Gamma} \|f\|_{L^\infty}^2 \|f_g\|^2 \\ &= \|f\|_{L^\infty}^2 \left\| \sum_{g \in \Gamma} \delta_g \otimes f_g \right\|^2. \end{aligned}$$

Therefore, the operator $\pi_\alpha(f)$ is bounded with norm at most $\|f\|_{L^\infty}$. Furthermore, by applying $\pi_\alpha(f)$ to $\delta_g \otimes 1$, it follows that $\|\pi_\alpha(f)\| = \|f\|_{L^\infty}$.

- (b) We start by using the fact that $\alpha_{g^{-1}}$ is linear and the bilinearity of the tensor product to show that π_α is linear.

Next, observe that

$$\begin{aligned} \pi_\alpha(fh)(\delta_g \otimes f_g) &= \delta_g \otimes \alpha_{g^{-1}}(fh)f_g \\ &= \delta_g \otimes \alpha_{g^{-1}}(hf) \end{aligned}$$

- (c) We recall that the action of λ_g on $\ell^2(\Gamma) \otimes L^2(X, \mu)$ is given on simple tensors by

$$\lambda_g(\delta_h \otimes f_h) = \delta_{gh} \otimes f_h.$$

Therefore, computing, we have

$$\begin{aligned} \lambda_g \pi_\alpha(f) \lambda_{g^{-1}}(\delta_h \otimes f_h) &= \lambda_g \pi_\alpha(f)(\delta_{g^{-1}h} \otimes f_h) \\ &= \lambda_g(\delta_{g^{-1}h} \otimes \alpha_{h^{-1}g}(f)f_h) \\ &= \delta_h \otimes \alpha_{h^{-1}}(\alpha_g(f))f_h \\ &= \pi_\alpha(\alpha_g(f))(\delta_h \otimes f_h). \end{aligned}$$

Problem (Problem 2): Suppose $\Gamma \curvearrowright^\alpha L^\infty(X, \mu)$ for a countable discrete group Γ and a probability space (X, μ) . For $f \in L^\infty(X, \mu)$, define $\phi_\alpha(f) \in B(\ell^2(\Gamma) \otimes L^2(X, \mu))$ by

$$\phi_\alpha(f)(\delta_g \otimes f_g) = \delta_g \otimes f_g f,$$

and define $\rho(g)$ for $g \in \Gamma$ by

$$\rho(g)(\delta_h \otimes f_h) = \delta_{hg^{-1}} \otimes \alpha_g(f_h).$$

Show that $\phi_\alpha(L^\infty(X, \mu)) \cup \rho(\Gamma) \subseteq (L^\infty(X, \mu) \rtimes_\alpha \Gamma)'$.

Solution: We will compute directly on simple tensors as follows:

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$$\begin{aligned} \lambda_g \rho_h(\delta_k \otimes f_k) &= \lambda_g(\delta_{kh^{-1}} \otimes \alpha_h(f_k)) \\ &= \delta_{gkh^{-1}} \otimes \alpha_h(f_k) \\ &= \rho_h(\delta_{gk} \otimes f_k) \\ &= \rho_h \lambda_g(\delta_k \otimes f_k). \end{aligned}$$

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$$\begin{aligned} \pi_\alpha(f) \phi_\alpha(g)(\delta_k \otimes f_k) &= \pi_\alpha(f)(\delta_k \otimes f_k g) \\ &= \delta_k \otimes \alpha_{k^{-1}}(f)(f_k g) \\ &= \delta_k \otimes (\alpha_{k^{-1}}(f) f_k) g \\ &= \phi_\alpha(g)(\delta_k \otimes \alpha_{k^{-1}}(f) f_k) \\ &= \phi_\alpha(g) \pi_\alpha(f)(\delta_k \otimes f_k). \end{aligned}$$

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$$\begin{aligned} \pi_\alpha(f) \rho_g(\delta_k \otimes f_k) &= \pi_\alpha(f)(\delta_{kg^{-1}} \otimes \alpha_g(f_k)) \\ &= \delta_{kg^{-1}} \otimes \alpha_{gk^{-1}}(f) \alpha_g(f_k) \\ &= \delta_{kg^{-1}} \otimes \alpha_g(\alpha_{k^{-1}}(f) f_k) \\ &= \rho_g(\delta_k \otimes \alpha_{k^{-1}}(f) f_k) \\ &= \rho_g \pi_\alpha(f)(\delta_k \otimes f_k). \end{aligned}$$

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$$\begin{aligned} \phi_\alpha(f) \lambda_g(\delta_k \otimes f_k) &= \phi_\alpha(f)(\delta_{gk} \otimes f_k) \\ &= \delta_{gk} \otimes f_k f \\ &= \lambda_g(\delta_k \otimes f_k) \\ &= \lambda_g \phi_\alpha(f)(\delta_k \otimes f_k). \end{aligned}$$

Problem (Problem 3): Suppose $\Gamma \curvearrowright^\alpha L^\infty(X, \mu)$.

- (a) Show that $\delta_e \otimes 1$ is a cyclic vector for $L^\infty(X, \mu) \rtimes_\alpha \Gamma$.
- (b) Show that $\delta_e \otimes 1$ is a separating vector for $L^\infty(X, \mu) \rtimes_\alpha \Gamma$.

Solution:

- (a) We start by observing that if $E \subseteq X$ is a measurable subset, and $g \in \Gamma$ is an arbitrary group element,

then

$$\lambda_g \pi_\alpha(\alpha_g(\chi_E))(\delta_e \otimes 1) = \delta_g \otimes \chi_E.$$

By taking finite linear combinations of elements of this form, we obtain all elements of the tensor product $\ell^2(\Gamma) \otimes L^2(X, \mu)$ given by $\delta_g \otimes \theta$, where θ is a simple function. Since elements of this form are dense in $\ell^2(\Gamma) \otimes L^2(X, \mu)$, it follows that $\delta_e \otimes 1$ is a cyclic vector for $L^\infty(X, \mu) \rtimes_\alpha \Gamma$.

- (b) A vector is separating for a von Neumann algebra M if and only if it is cyclic for the commutant M' . From our work in Problem 2, we know that the commutant of $L^\infty(X, \mu) \rtimes_\alpha \Gamma$ contains the von Neumann algebra generated by $\phi_\alpha(L^\infty(X, \mu))$ and $\rho(\Gamma)$.

Observe then that if $E \subseteq X$ is measurable and $g \in G$, we have

$$\rho_g \phi_\alpha(\alpha_{g^{-1}}(\chi_E))(\delta_e \otimes 1) = \delta_g \otimes \chi_E.$$

Therefore, by the same reasoning as in part (a), it follows that $\delta_e \otimes 1$ is cyclic for $(L^\infty(X, \mu) \rtimes_\alpha \Gamma)'$, so $\delta_e \otimes 1$ is separating for $L^\infty(X, \mu) \rtimes_\alpha \Gamma$.